



Deep learning assisted sound source localization from a flying drone

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OVERVIEW

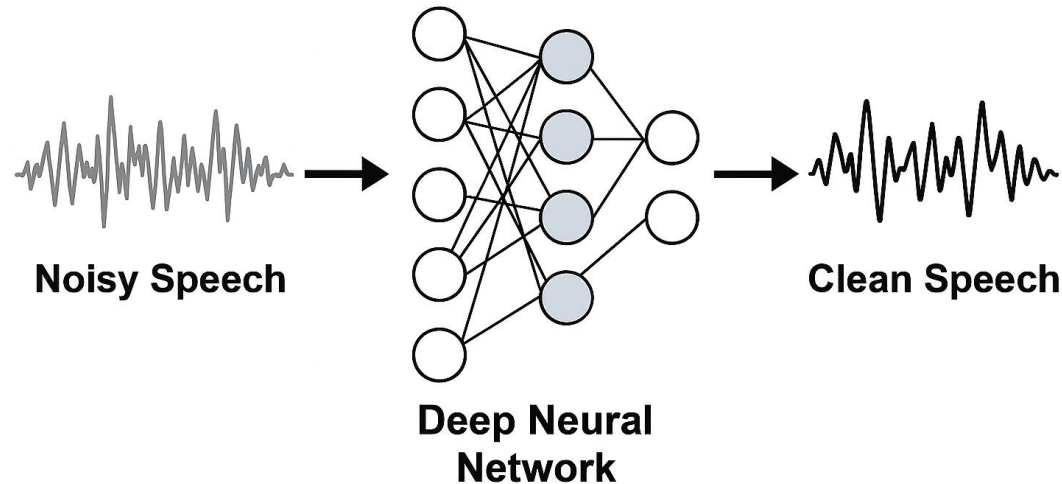
- ① Introduction
- ② Motivation
- ③ Brief discussion about Drone noise
- ④ Experiment testbed
- ⑤ Implementation video demo
- ⑥ Methodology
- ⑦ Validation
- ⑧ Conclusion
- ⑨ Limitation
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- ⑪ Question

INTRODUCTION

- Research goal
 - ➡ Develop a deep learning assisted framework
 - ➡ Suppress ego-noise and localize sound sources
- Core Problem
 - ➡ Severe ego-noise from drone rotors and propellers
 - ➡ Movement of both drone and source
- Context and Scope
 - ➡ Drone audition to detect, enhance and understand sounds
 - ➡ Useful for search, rescue monitoring and human interaction

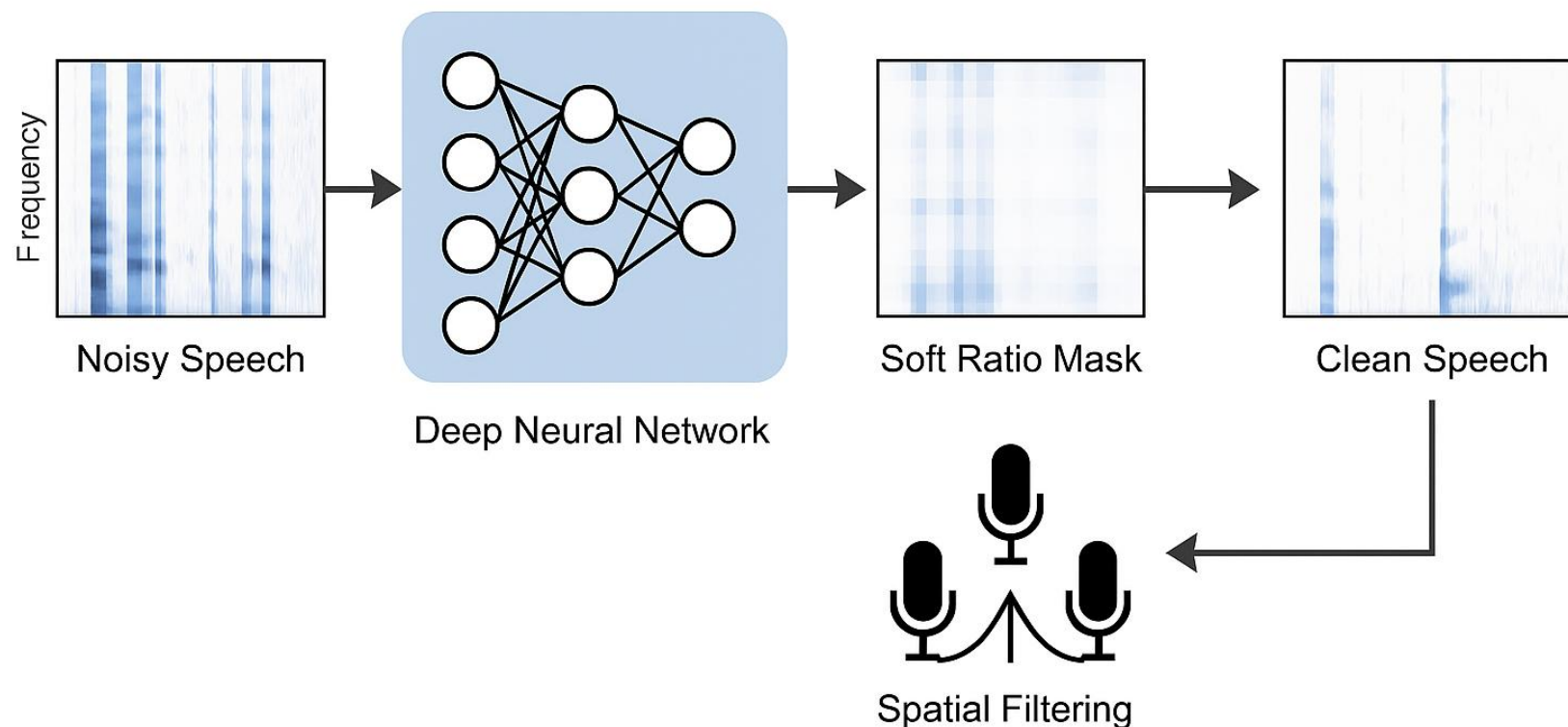
MOTIVATION

- **Conventional methods fails** under drone noise conditions
 - ➔ Steered response power with phase transform (SRP-PHAT)
 - ➔ Multiple signal classification (MUSIC)
 - ➔ Time-frequency spatial filtering (TFS)
- **Deep neural network** for speech enhancement in low SNR



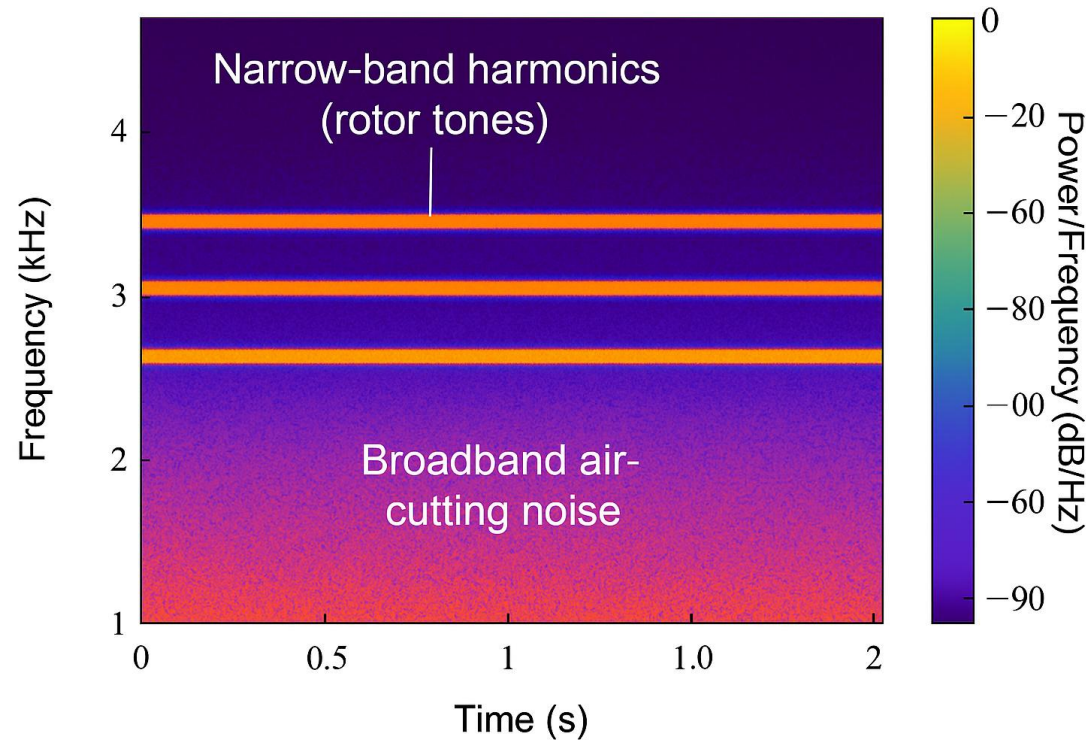
MOTIVATION (cont..)

- Integrating DNN mask estimation with spatial filtering



CHALLENGES

- Ego-Noise Characteristics
 - ➔ Narrow-band harmonics (rotor tones) + broadband air-cutting noise.
- Low Signal-to-Noise Ratios
 - ➔ Often below -15 dB.



CHALLENGES (Cont..)

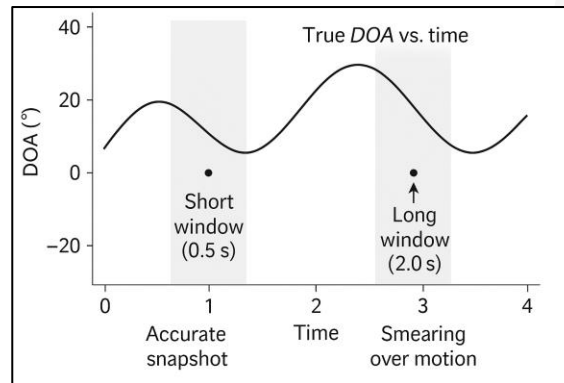
○ Dynamic Acoustics

➡ Both drone and source move continuously.



○ Limited Processing Window

➡ Localization must work with short segments (≈ 0.5 s).



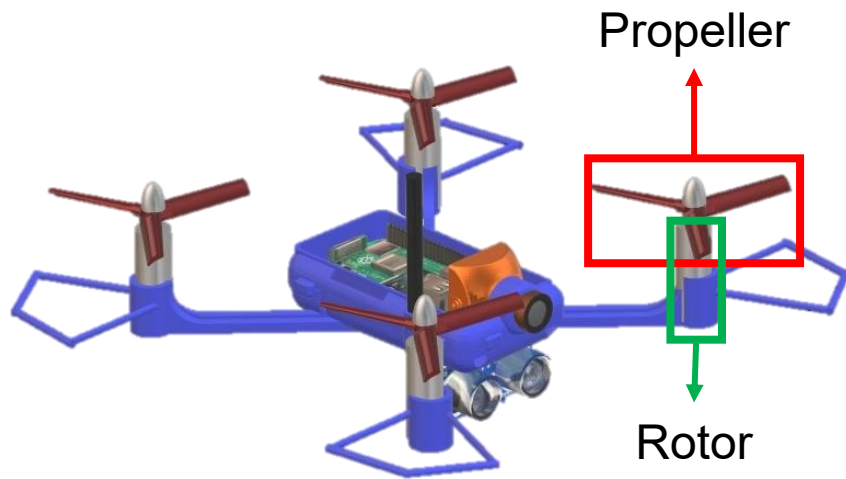
CHALLENGES (Cont..)

- Hardware Constraints

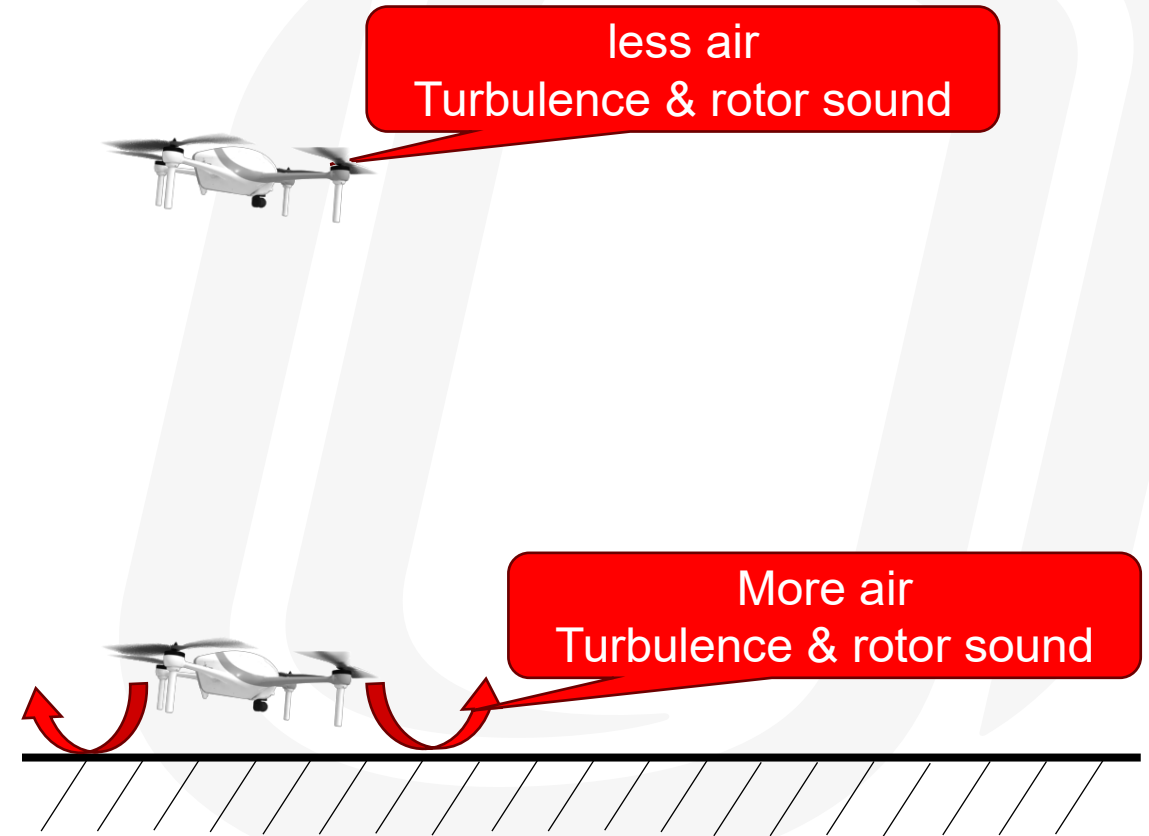
- ➡ Microphone arrays on small UAVs should be light

- ➡ The rotors and arms create continuous vibration and air turbulence.

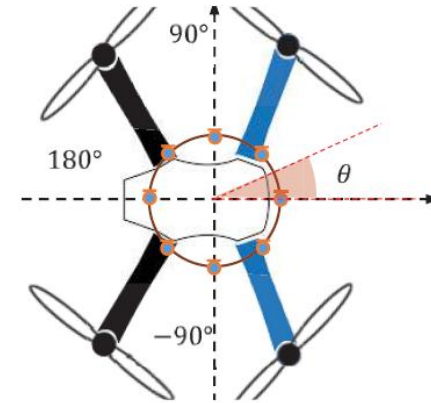
BRIEF DISCUSSION ABOUT DRONE NOISE



Ego noise source in a drone



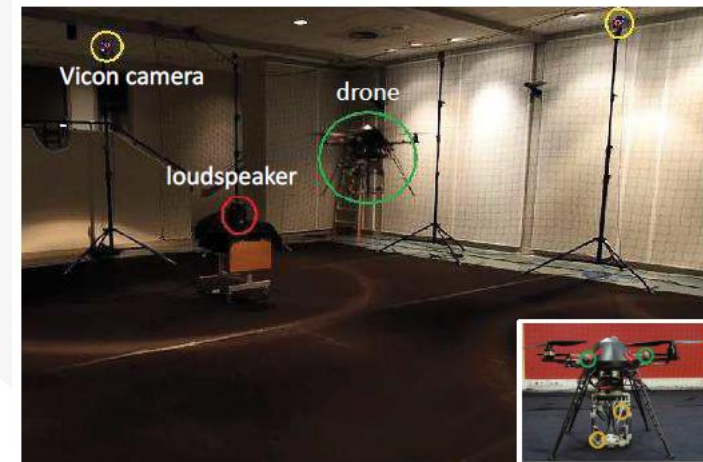
EXPERIMENT TESTBED



The 2D coordinate system for a circular array mounted on a AVQ drone.

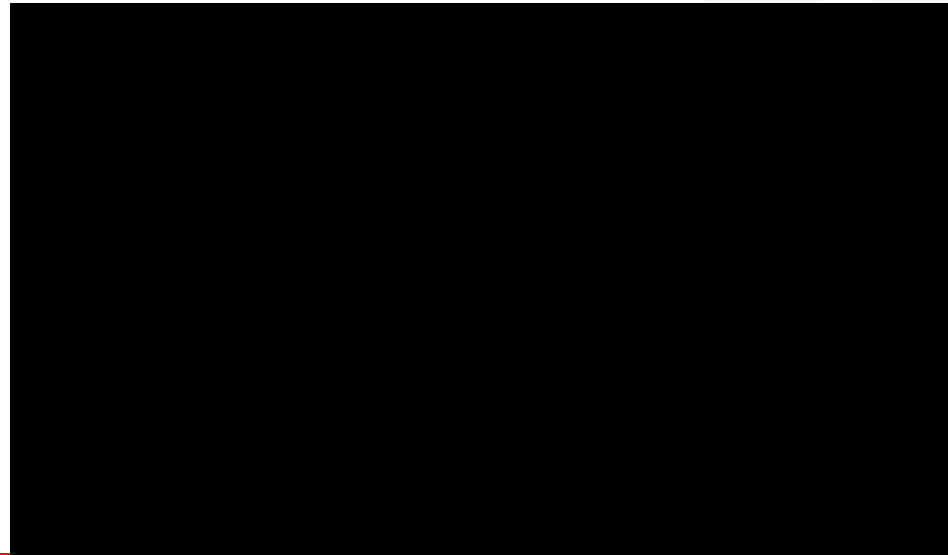


A static drone with a moving speaker



A flying drone with a static speaker

IMPLEMENTATION VIDEO DEMO



METHODOLOGY

First step – Mask estimation of useful sound and enhance the sound

⇒ microphone signal in the time domain

$$x(n) = s(n) + v(n)$$

⇒ Time-frequency domain as,

$$X(k, l) = S(k, l) + V(k, l)$$

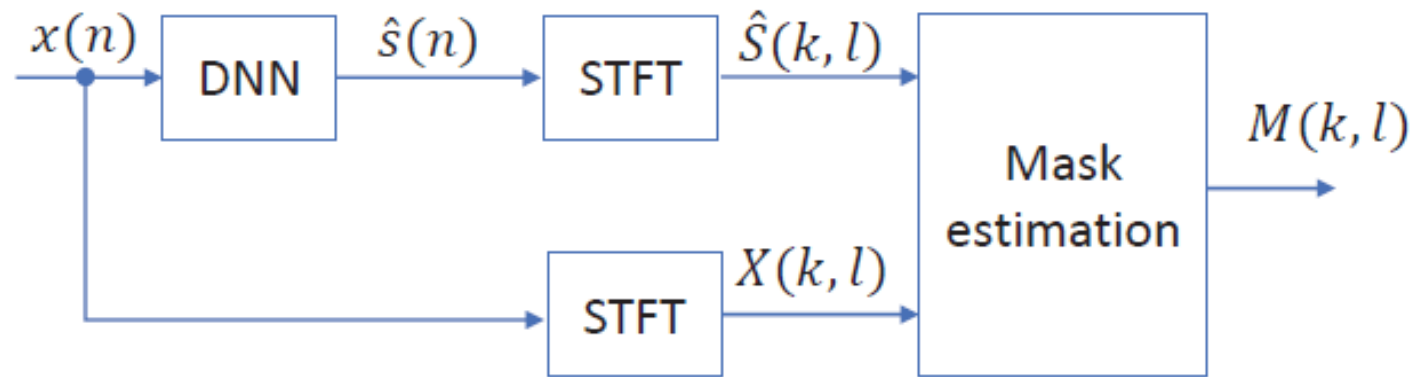


Fig: Single-channel sound enhancement DNN model that estimates a soft ratio mask

METHODOLOGY (Cont..)

Second step – multi-channel sound source localization pipelines

- ① SRP-DNN employs time-frequency weighting when computing the spatial likelihood map.
- ② TFS-DNN employs time-frequency weighting when computing the target correlation matrix and the spatial filter.

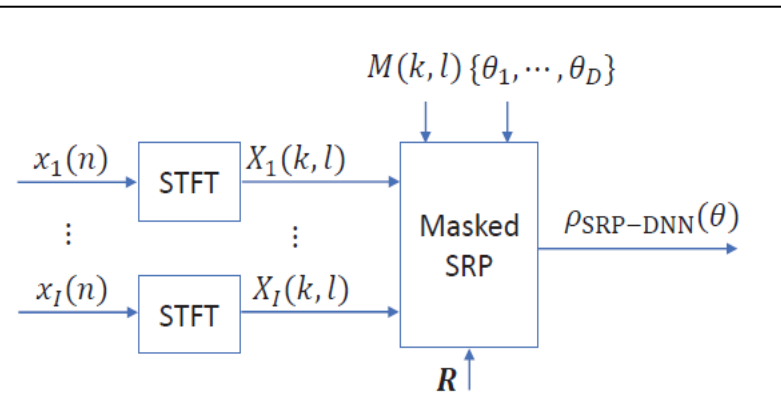


Fig: SRP-DNN

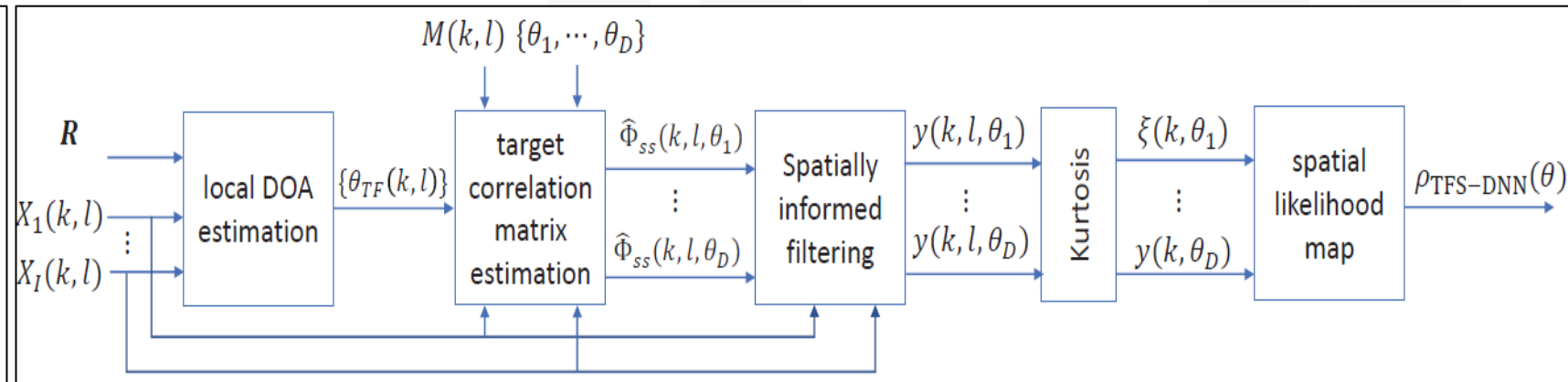
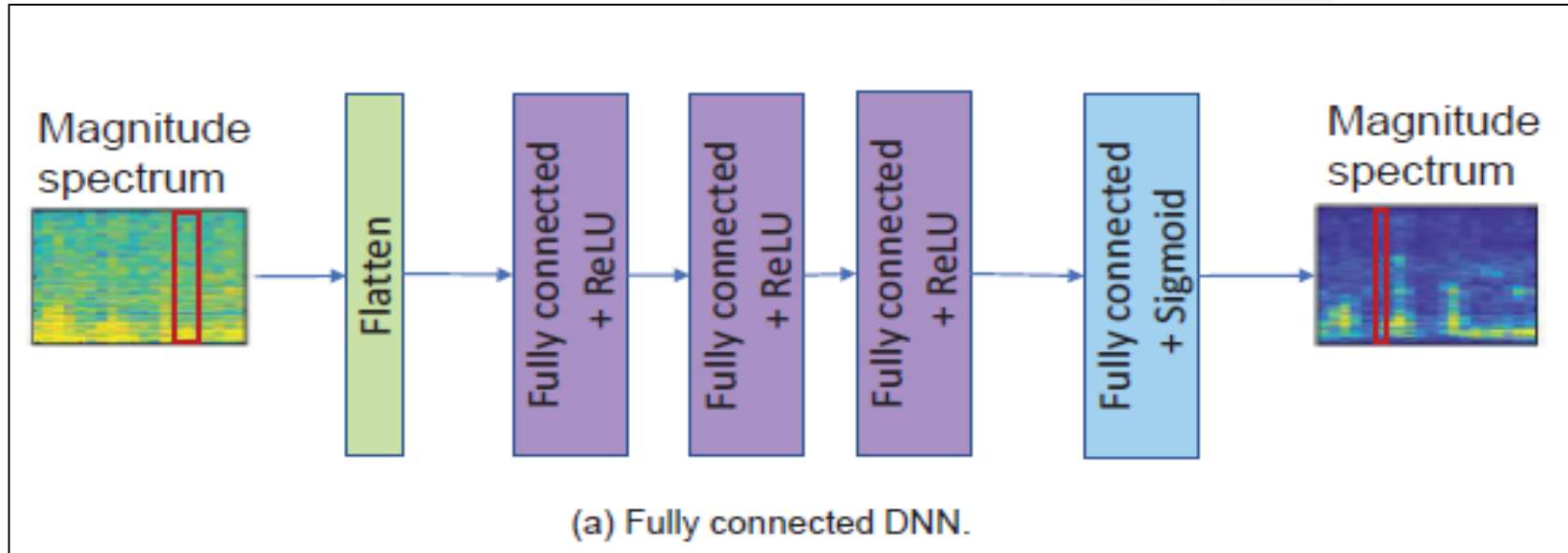


Fig: TFS-DNN

METHODOLOGY (Cont..)

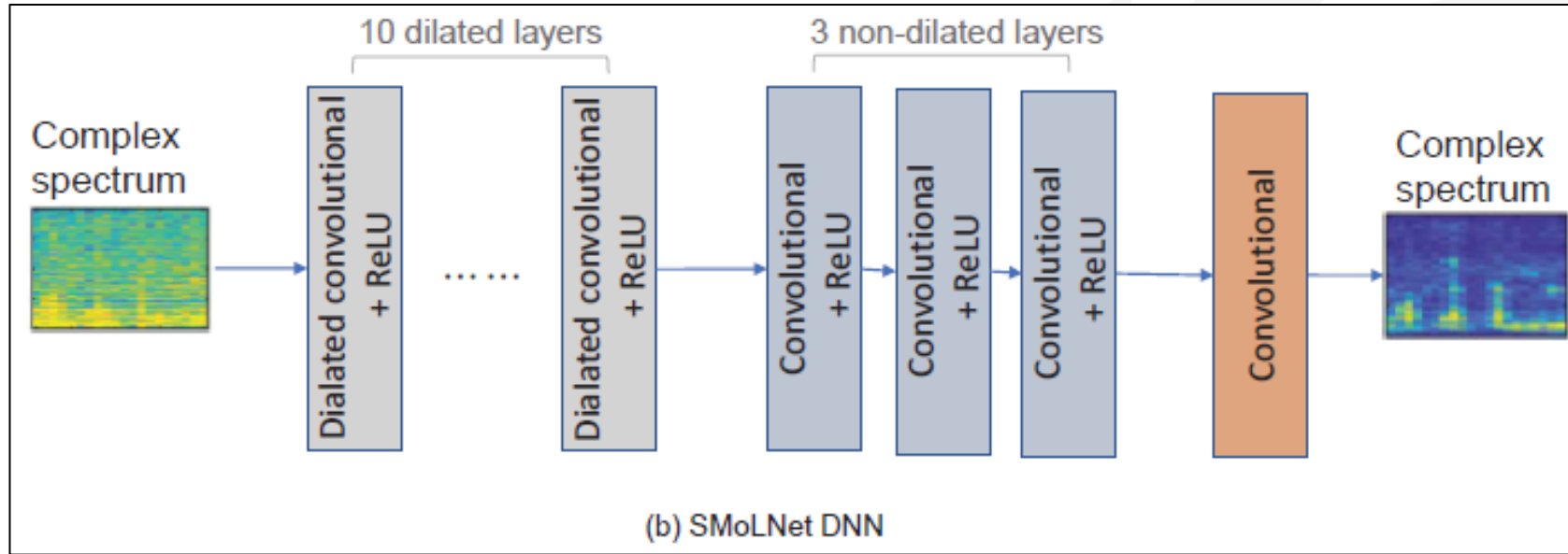
Mask Estimation by DNN Sound Enhancement



- ★ Operates in time-frequency magnitude domain.
- ★ Convert the time domain signal into STFT domain then this input to DNN.
- ★ One flatten layer, three fully connected layers and one output layer.
- ★ Each fully connected layers contain 2048 neurons.
- ★ Output layer generate STFT frame of magnitude estimation.

METHODOLOGY (Cont..)

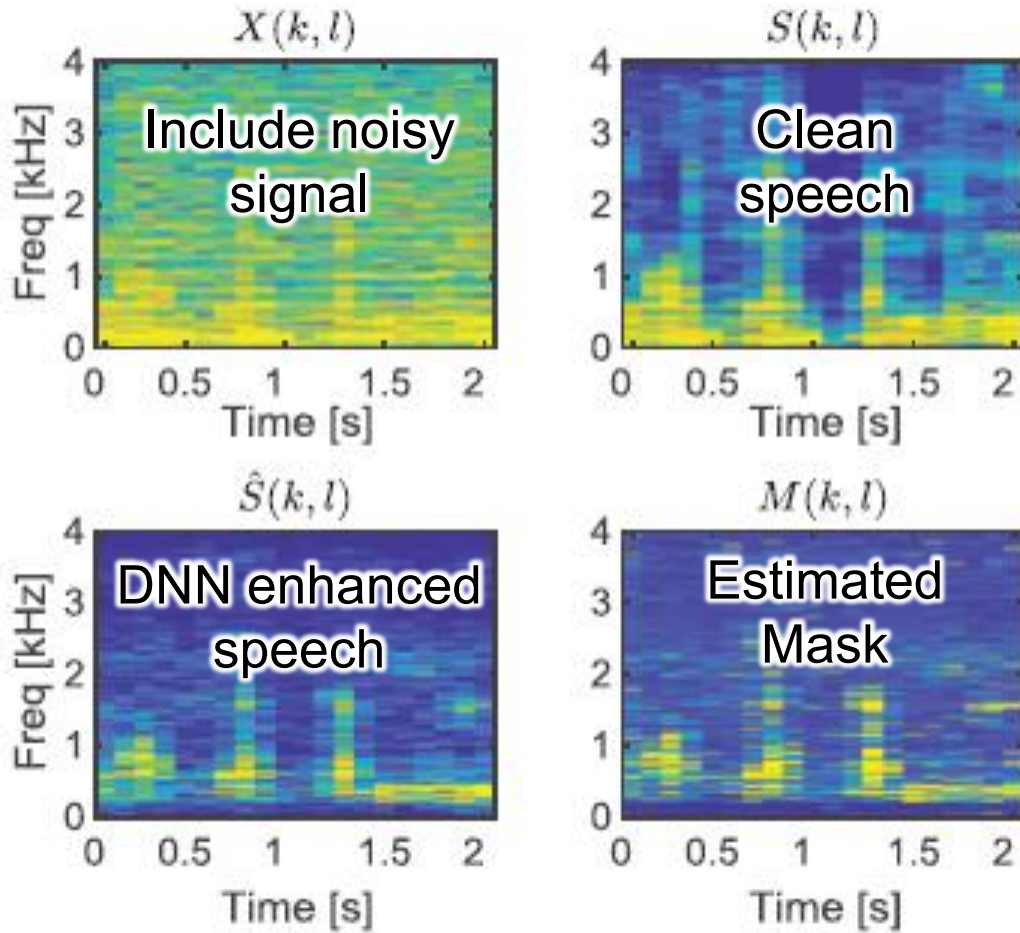
Mask Estimation by DNN Sound Enhancement



- ★ Operates in time-frequency complex domain.
- ★ Ten dilated convolutional layers, three non-dilated convolutional layers and one output layer.
- ★ SMoLNet takes as input the whole STFT complex spectrum, and generates a complex spectrum of the same size, which can be converted to the time domain as clean speech estimation.

METHODOLOGY (Cont..)

Mask Estimation Result



- ★ Used SMoLNet as DNN sound enhancement model.
- ★ Input SNR -5dB, and it was enhanced by 11dB
- ★ Estimated mask $M(k, l)$ appears consistent with the energy distribution of the clean speech $S(k, l)$ in the time-frequency domain

METHODOLOGY (Cont..)

Sound Source Localization

SRP-DNN

① Multi-microphone signals & STFT of each microphone signal $X_m(k,l)$

② Compute cross-correlations between microphones & normalize. $\frac{X_{m_1}^*(k,l)X_{m_2}(k,l)}{|X_{m_1}(k,l)X_{m_2}(k,l)|}$

③ Apply phase delay compensation for a candidate direction θ

④ Sum of all frequencies and time frames.

⑤ Estimate source location

TFS-DNN

① Compute instantaneous spatial likelihood at each time–frequency bin

② Confidence measures with DNN mask for each candidate direction θ

$$\tilde{c}_d(k,l,\theta) = c_d(k,l,\theta) \cdot M(k,l)$$

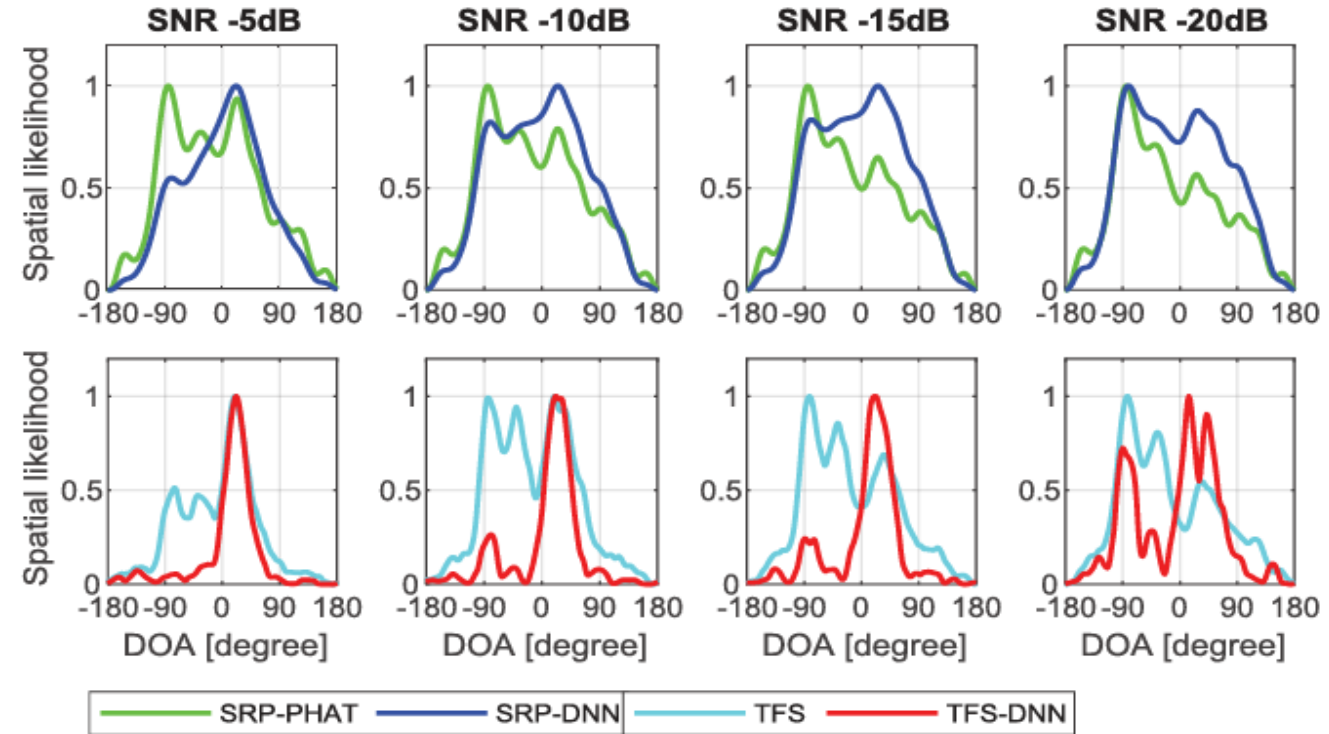
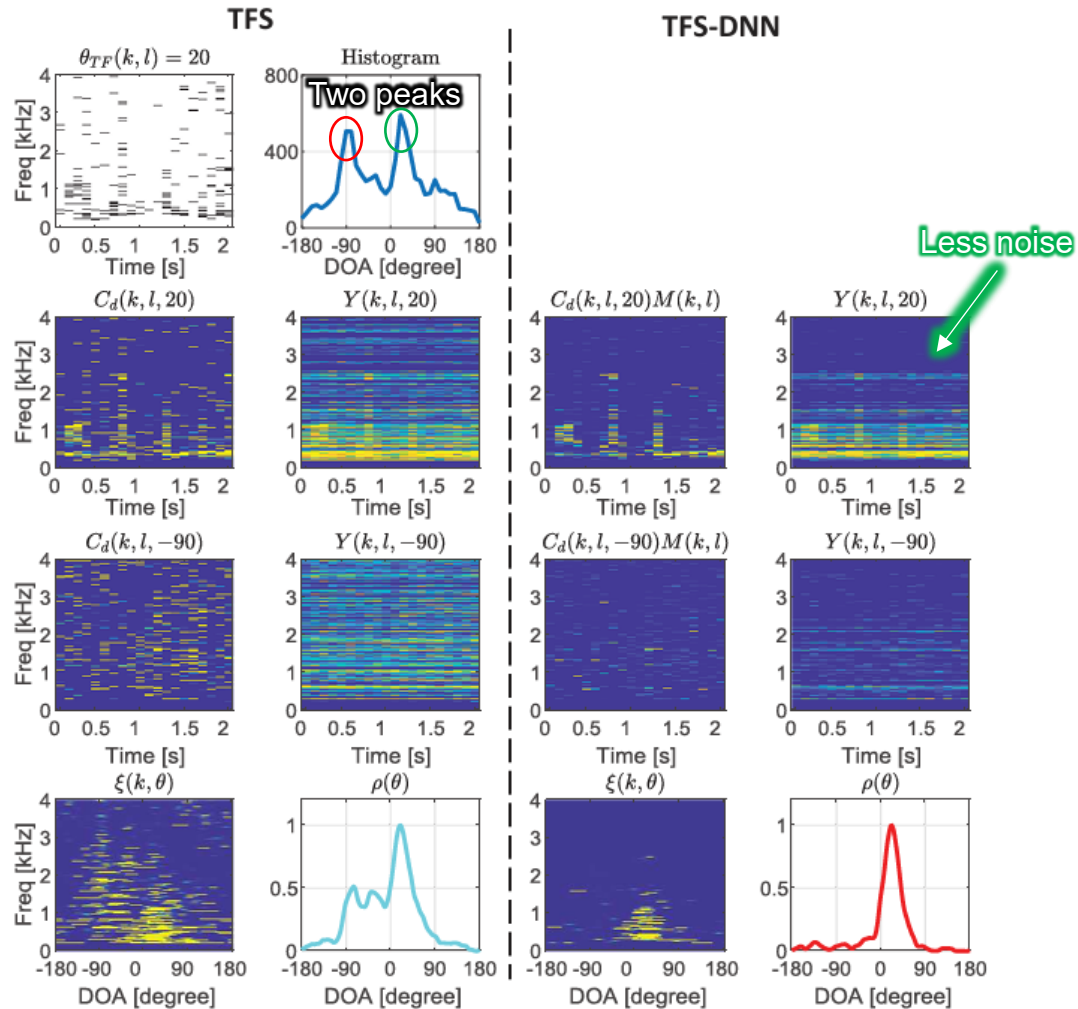
③ Target Correlation Matrix form the spatial covariance matrix

④ Spatial Filtering (Beamforming)

⑤ Kurtosis-based Spatial Likelihood

⑥ Source DOA Estimation

METHODOLOGY (Cont..)



- Overall, the DNN-estimated ratio mask can indicate the speech presence probability at each time-frequency bin.
- As a result, SRP-DNN and TFS-DNN work more robustly in low-SNR scenarios than their counterparts SRP-PHAT and TFS, respectively.

VALIDATION

Consider eight algorithms for comparisons

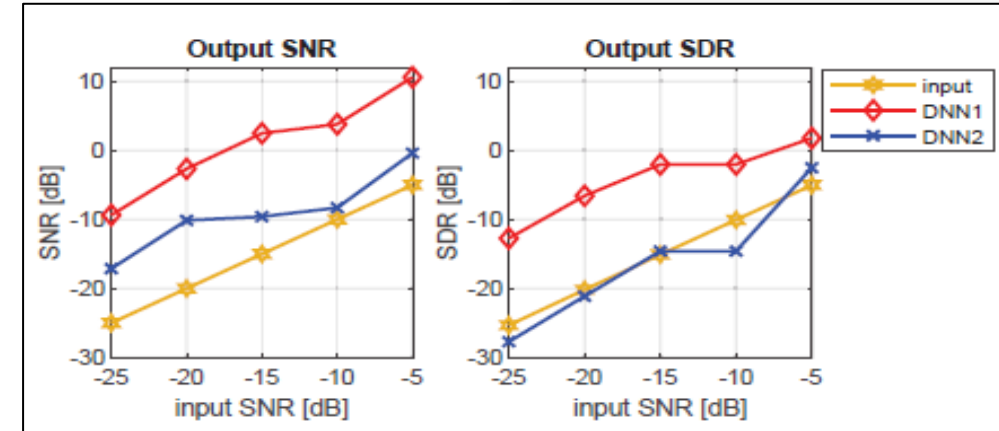
Two base line approaches

1. SRP-PHAT
2. TFS

Six DNN base approaches

1. SRP-DNN0
2. SRP-DNN1
3. SRP-DNN2
4. TFS-DNN0
5. TFS-DNN1
6. TFS-DNN2

★ Here DNN0 is clean speech reference
DNN1 is SMLoNet and DNN2 is FC



The output SNR and SDR achieved by the DNN1 (SMoLNet) and DNN2 (FC) at an input SNR varying from -25 to -5 dB.

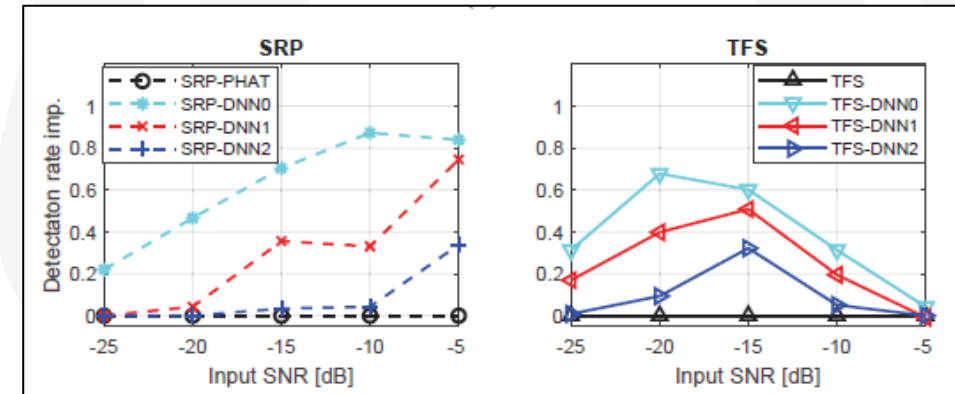


Figure shows the detection rate improvement by using DNN over SRP and TFS at block size 1s and various input SNRs.

CONCLUSION

1. TFS-DNN successfully integrates deep learning based spectral masking with time frequency spatial filtering, achieving accurate localization even under extremely low SNRs (-20 dB).
2. DNN preprocessing effectively reduces drone ego-noise and enhances traditional spatial methods such as SRP and TFS.
3. The framework generalizes well across different datasets involving static, moving sources and flying drones.
4. Demonstrates strong localization performance with short processing windows (~ 0.5 s), essential for real-time UAV audition.

LIMITATION

1. TFS-DNN requires repeated spatial filter computation across angles and frequencies not yet suitable for real-time onboard processing on small UAV CPUs.
2. Experiments use laboratory datasets, but real-world outdoor conditions (wind turbulence, reverberation) remain untested.
3. The algorithm targets single-source localization; multi-source or overlapping sounds weren't explicitly evaluated.
4. Last but not the least, there was no security concern with the DNN models.



Now what we are doing in ASTRA Lab ?

DETECT GUNSHOT

Datasets

15.Smith&Wesson38special
16.SportKing22
17.WASR10
18.WinchesterM14

- 18 class of guns
- 1,781 audio for training
- Pretrained model of 2 million YouTube videos and contains 527 classes.

11.Ruger22
12.Ruger357
13.Sig9
14.Smith&Wesson22

DNN for Gunshot classification

Spectrogram Extr

Logmel_extractor

BatchNorm2d

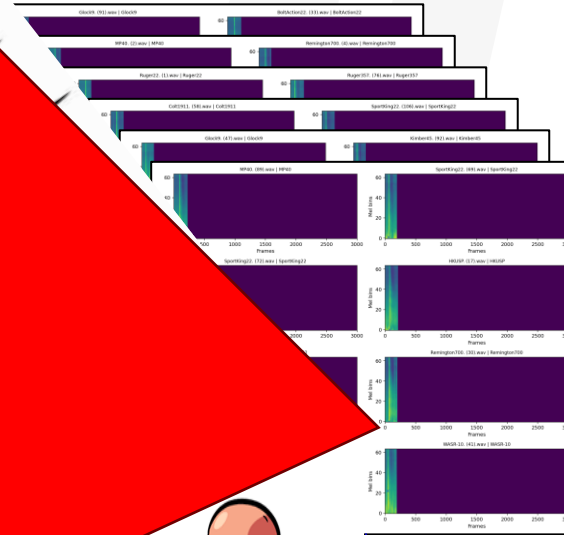
Conv_block1
Conv1 : Conv2d
Conv2 : Conv2d
Bn1, Bn2

Conv_block6

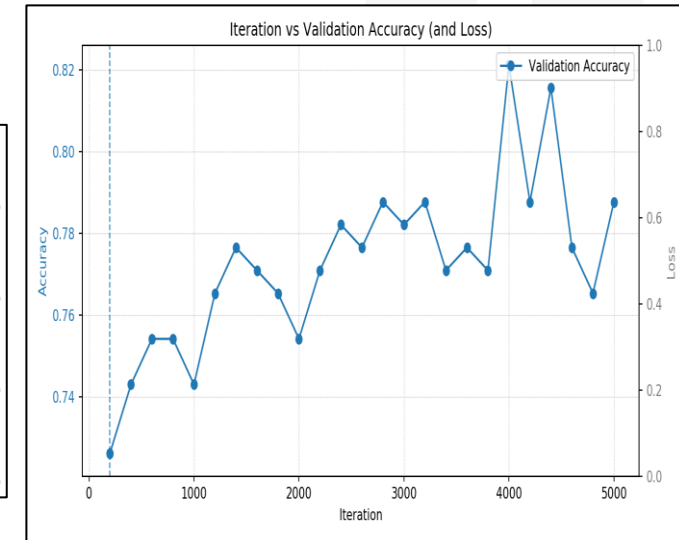
Fc1

fc_transfer

Input for training samples



Training result



Questions

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